THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
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ANSWER KEY & MARKING SCHEME SET A

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 4 g/cm³	Density = mass ÷ volume
2	**(d)** uniform mixture	Same composition throughout
3	**(c)** unsaturated	At that temperature
4	**(b)** decreases	Fizz escapes from warm soda
5	**(d)** space occupied by an object	Measured in mL or cm³
6	**(b)** mass per unit volume	ρ = m/V
7	**(b)** more than water	Heavier per volume
8	**(d)** decreases	Substances expand
9	**(d)** a homogeneous mixture	Uniform throughout
10	**(c)** 30 g	250 − 220

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A uniform mixture of two or more substances (1). Salt in water (1).	1+1
12	Solute is the dissolved substance, smaller amount (1). Solvent is the medium, larger amount (1).	1+1
13	Saturated: no more solute dissolves at that temperature (1). Unsaturated: more solute can dissolve (1).	1+1
14	Maximum amount of solute that dissolves in a fixed quantity of solvent (1). At a given temperature (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Add solute in small portions to water and stir (1). When no more dissolves, it is saturated (1). Undissolved solute settles at the bottom (1).	1+1+1
16	Add sugar to cold water until saturated (1). Warm the water and more sugar dissolves (1). So solubility increases with temperature (1).	1+1+1
17	Solubility = 36 g per 100 mL (1½). 10 g stays undissolved (1½).	1½+1½
	OR: Saturated: maximum dissolved (1). Unsaturated: can dissolve more (1). Dilute: small solute amount (1).	1+1+1
18	Substance expands on heating (1½). Volume increases while mass is the same (1½).	1½+1½
	OR: Density = 120 ÷ 40 = 3 g/cm³ (1½). More than 1 g/cm³ so it sinks (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Solute and solvent (1). Solution (1). Saturated (1). Unsaturated (1). Solubility (1).	1+1+1+1+1
	OR: Take 100 mL water (1). Add salt in portions with stirring (1). Stop when it just stays undissolved (1). Weigh the salt added (1). That mass is the solubility (1).	1+1+1+1+1
20	Solids increase (1). Sugar, salt examples (1). Gases decrease (1). Soda and dissolved oxygen examples (1). Importance (1).	1+1+1+1+1
	OR: Definition (1). Find mass with balance (1). Find volume by displacement or measuring (1). Divide mass by volume (1). Record with unit (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 2.7 g/cm³	Density = mass ÷ volume
2	**(b)** salt	Dissolved substance
3	**(d)** saturated	Maximum amount dissolved
4	**(a)** gas dissolves better in cold liquid	Gas solubility falls with temperature
5	**(b)** weighing balance	Instrument for mass
6	**(a)** kg/m³	Also g/cm³ in practice
7	**(a)** lower density than water	Particles are farther apart
8	**(c)** gases	Gases compress easily
9	**(b)** salt in water	Solid dissolved in liquid
10	**(c)** more solute is added until it stops dissolving	Max reached

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Solubility of most solids increases (1). That of gases decreases (1).	1+1
12	Curved surface of a liquid in a measuring cylinder (1). Read at eye level at the bottom of the curve (1).	1+1
13	Mass per unit volume (1). Density = mass ÷ volume (1).	1+1
14	kg/m³ (1). g/cm³ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Wood floats (1). Iron sinks (1). Because of density compared with water (1).	1+1+1
16	Gases compress easily so density rises (1½). Liquids hardly compress so density changes little (1½).	1½+1½
17	Fill cylinder with water and note volume (1). Lower stone gently (1). Difference in readings is the stone's volume (1).	1+1+1
	OR: Salt solution has higher density (1½). Egg is less dense than that solution (1½).	1½+1½
18	Water particles rearrange on freezing (1). They occupy more space (1). So ice has lower density (1).	1+1+1
	OR: Stirring (1). Temperature (1). Size of solute particles (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Ice particles are farther apart (1). Ice has lower density (1). Floats and forms top layer (1). Insulates water beneath (1). Aquatic life survives (1).	1+1+1+1+1
	OR: Temperature rise lowers density (1). Example – hot air rises (1). Pressure raises gas density (1). Liquids and solids change little (1). Example – compressed gas cylinder (1).	1+1+1+1+1
20	(i) 3 g/cm³ (2). (ii) Sinks since density is more than water (1). (iii) 150 g (2).	2+1+2
	OR: Place egg in tap water, it sinks (1). Add salt gradually with stirring (1). Water density increases (1). Egg floats when solution is denser (1). Sinking again on dilution (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.